

# Ayush Maurya

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## SUMMARY

Software engineer with full-stack development experience building REST APIs and database-backed applications using Python, JavaScript, MongoDB, and React. Published open-source Python package (ElectoralSim) on PyPI. Experienced in system design, API development, Git version control, database design, algorithm design, and cloud deployment. Built real-time IoT monitoring dashboard (MERN stack), LLM-integrated client applications, and custom full-stack web platforms. Co-founded startup (CrowdTwin) simulation architecture for crowd dynamics.

## EDUCATION

|                                                                              |              |
|------------------------------------------------------------------------------|--------------|
| IIIT Hyderabad                                                               | 2021–Present |
| B.Tech in Computer Science, M.S. by Research in Computing and Human Sciences |              |
| Dual degree offered by the Human Sciences Research Center                    |              |
| Mary Gardiner's Convent School                                               | 2021         |
| Class XII — 90.1%                                                            |              |
| City Montessori School                                                       | 2019         |
| Class X — 94.2%                                                              |              |

## EXPERIENCE

|                                                                                                                                                                                                                                 |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| SERC Digital Twin                                                                                                                                                                                                               | Environmental IoT |
| <ul style="list-style-type: none"><li>Built real-time MERN dashboard streaming IoT data from 4 environmental sensors.</li><li>Monitored temperature, humidity, water, and air quality through live visualizations.</li></ul>    |                   |
| TAFEA                                                                                                                                                                                                                           | System Design     |
| <ul style="list-style-type: none"><li>Engineered student client integrating LLM APIs through structured SDLC methodology.</li><li>Designed tailored student workflows with AI-powered features and iterative testing.</li></ul> |                   |

## PROJECTS

|                                                                                                                                                                                                                                    |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| ElectoralSim                                                                                                                                                                                                                       | PyPI Package         |
| <ul style="list-style-type: none"><li>Published vectorized Python library on PyPI for electoral system simulation.</li><li>Architected simulation engine supporting multiple voting methods and district configurations.</li></ul> |                      |
| CrowdTwin                                                                                                                                                                                                                          | Startup Co-Founder   |
| <ul style="list-style-type: none"><li>Created crowd simulation architecture for major event planning as co-founder.</li><li>Built predictive dashboard prototype for real-time crowd behavior analytics.</li></ul>                 |                      |
| ayushmaurya.online                                                                                                                                                                                                                 | Full-Stack Portfolio |
| <ul style="list-style-type: none"><li>Built full-stack portfolio on custom Ubuntu stack without framework dependencies.</li><li>Deployed Linux server environment handling DNS, SSL, and reverse proxy.</li></ul>                  |                      |
| ML for Local Geoguesser                                                                                                                                                                                                            | Deep Learning        |
| <ul style="list-style-type: none"><li>Built ResNet50 model achieving 92% region classification accuracy.</li></ul>                                                                                                                 |                      |
| Highland History Project                                                                                                                                                                                                           | OCR + LLM Pipeline   |
| <ul style="list-style-type: none"><li>Developed OCR and LLM pipeline for historical document digitization.</li></ul>                                                                                                               |                      |

## TECHNICAL SKILLS

**Languages:** Python, C/C++, JavaScript, SQL, HTML/CSS  
**Technologies:** Machine Learning, PyTorch, Linux, System Design, Systems Programming  
**UI/UX and Design:** Wireframing, Prototyping, User Stories, User Research, Interaction Design

## HONOURS

UGEE Rank: 111 out of 40,000+ candidates  
JEE Main: 96.9 percentile  
School Grades: Class XII — 90.1%; Class X — 94.2%

## ACTIVITIES

Bass guitarist in college band and lawn tennis player.